Lecture Scribe L17

L15_L17.pdf
PDF

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Use ONLY the attached lecture slides / lecture PDF as context. Do NOT use any textbook or external knowledge.

The scribe must be a faithful reconstruction of what was taught in class. Every definition, notation, assumption, result, proof step, derivation step, and example step must be directly traceable to the provided slides.

While writing the scribe, explicitly reason step by step through:

- Definitions and notation
- Assumptions and conditions
- Statements of results or theorems
- Proofs or derivations (only where they appear in the slides)
- Worked examples (only where they appear in the slides), including all intermediate steps

Make all logical dependencies between concepts explicit so that a student can reliably revise using only this document.

Follow the lecture's original structure and scope. Preserve ordering of topics as presented.

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STRICT CONSTRAINTS:

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- Do NOT add new examples, proofs, or derivations.
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This output is a lecture scribe for exam preparation, not a tutorial, summary, or creative explanation.

Lecture Scribe: Joint Random Variables, Joint Distributions, and Examples

This scribe is reconstructed strictly from the attached lecture slides.

L15_L17

1. Why Study Joint Random Variables?

In science and in real life, we are often interested in **two (or more) random variables at the same time**.
L15_L17

Examples listed in the lecture

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members
L15_L17

Example #1: Smart Transportation

In a smart traffic system, define:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question posed in the lecture:

- Can speed be high when the traffic density is also high?

Answer shown:

- Yes / No

Hence, X and Y are presented as **correlated random variables**.

Real ITS applications listed:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing
L15_L17

Example #2: Wireless Network Performance

In a wireless system:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Reason for joint random variables:

- Throughput depends on SNR
- Both vary due to fading, interference, and congestion

Question posed in the lecture:

- If SNR is high, will throughput always be high?

Answer shown:

- Yes / No

Reasoning stated:

- Network Congestion (condition relationships)

Network applications listed:

- 5G scheduling
- Adaptive modulation
- Network planning

L15_L17

Example #3

- Weather Prediction: $X =$ Rainfall and $Y =$ Temperature
- Application: weather forecasting

L15_L17

Example #4

- Drone Delivery: $X =$ Wind Speed and $Y =$ Delivery Time
- Applications: logistics and UAV systems

L15_L17

Example #5

- Roll two dice $(X \text{ and } Y)$ using phones
- Events shown:
  - $X + Y = 7$
  - $X > Y$
- Statement: This requires joint PMF

L15_L17

Concluding statements from this section

In such situations, the random variables have a **joint distribution** that allows us to compute probabilities of events involving both variables and understand the **relationship** between the variables.

This is simplest when the variables are **independent**.

When they are not independent, we use **covariance** and **correlation** as measures of the nature of the dependence between them.

L15_L17

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2. Discrete Case — Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X \in \{x_1, x_2, \ldots, x_n\}, \qquad Y \in \{y_1, y_2, \ldots, y_m\}$$

Then the ordered pair (X, Y) takes values in the product set:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

L15_L17

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

This gives the probability of the joint outcome occurring simultaneously. L15_L17

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3. Joint PMF Table Representation

The joint PMF is organized in a table whose rows correspond to values $x_1, x_2, \ldots, x_n$ of X and whose columns correspond to values $y_1, y_2, \ldots, y_m$ of Y. The table entry in row x_i and column y_j is $p(x_i, y_j)$.

L15_L17

Key properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

L15_L17

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4. Example: Rolling Two Dice

Experiment

Roll two fair dice. L15_L17

Definitions

Let:

- $X =$ value on the first die
- $T =$ total (sum) of both dice

L15_L17

Since each outcome (i, j) has probability $\frac{1}{36}$, the lecture constructs the joint probability table of (X, T).

L15_L17

Joint Probability Table of (X, T)

Rows correspond to $X = 1, 2, 3, 4, 5, 6$, and columns correspond to $T = 2, 3, \ldots, 12$.

$X \backslash T$	2	3	4	5	6	7	8	9	10	11	12
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	0
2	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0
3	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0
4	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0
5	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0
6	0	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

L15_L17

Observations

- Entries are either 0 or $\frac{1}{36}$
- X and T are **not independent**

L15_L17

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5. Continuous Case — Joint PDF

The lecture states that the continuous case is analogous to the discrete case:

- Discrete values $\longrightarrow$ Continuous intervals
- Joint PMF $\longrightarrow$ Joint PDF
- Sums $\longrightarrow$ Double integrals

L15_L17

If

$$X \in [a, b], \qquad Y \in [c, d],$$

then the pair (X, Y) takes values in the product region:

$$[a, b] \times [c, d]$$

L15_L17

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** of X and Y if

$$P\big((X, Y) \text{ lies in a small rectangle of area } dx\,dy\big) \approx f(x, y)\,dx\,dy$$

L15_L17

The accompanying diagram shows a rectangle inside the region $[a, b] \times [c, d]$, with probability labeled as

$$\text{Prob.} = f(x, y)\,dx\,dy$$

for a small rectangle with side lengths dx and dy.
L15_L17

Key properties

$$f(x, y) \geq 0$$

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

L15_L17
The lecture also states:

- The joint PDF $f(x, y)$ may take values greater than 1.
- It is a probability density, not a probability.
L15_L17

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6. Worked Example for Joint PDF

Given

$$f(x, y) = 4xy, \qquad 0 \leq x \leq 1,\ 0 \leq y \leq 1$$

Tasks listed in the lecture:

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$

The lecture states that jointly, (X, Y) takes values in the **unit square**

$$[0, 1]^2$$

L15_L17
The figure on the slide shows the unit square, with the event A shaded in the region where $x < 0.5$ and $y > 0.5$.
L15_L17

Example Solution

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy\, dx\, dy = \int_0^1 \left(\int_0^1 4xy\, dx \right) dy = \int_0^1 2y\, dy = \left[y^2 \right]_0^1 = 1$$

Hence $f(x, y) \geq 0$ and total probability is 1. So it is a **valid joint PDF**.
L15_L17

Step 2: Compute $P(A)$

$$
\begin{aligned}
P(A) &= \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx \\
&= \int_0^{0.5} \left[2xy^2\right]_{0.5}^{1} \, dx \\
&= \int_0^{0.5} 2x(1 - 0.25) \, dx \\
&= \int_0^{0.5} \frac{3}{2} x \, dx \\
&= \frac{3}{16}
\end{aligned}
$$

L15_L17

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7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

L15_L17

The notation

$$X \leq x, \; Y \leq y$$

is used to mean the event

$$\{X \leq x \text{ and } Y \leq y\}$$

L15_L17

Definition

The joint cumulative distribution function is defined as

$$F(x, y) = P(X \leq x, \; Y \leq y)$$

L15_L17

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8. Joint CDF — Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, then the joint CDF is obtained by integrating the density:

$$F(x, y) = \int_c^y \int_a^x f(u, v) \, du \, dv$$

L15_L17

Recovering the Joint PDF

Since there are two variables, partial derivatives are used:

$$f(x,y) = \frac{\partial^2}{\partial x\, \partial y} F(x,y)$$

L15_L17

---

9. Joint CDF — Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then the joint CDF is obtained by summing probabilities:

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

L15_L17

Interpretation

- Add all probabilities in the table
- For rows with $x_i \le x$
- And columns with $y_j \le y$ L15_L17

---

10. Properties of the Joint CDF

The joint CDF $F(x,y)$ of X and Y must satisfy several properties. The slides list the following properties:

Property 1

$$F(x,y) \text{ is non-decreasing.}$$

If x or y increase, then $F(x,y)$ must stay constant or increase. L15_L17

Property 2

$$F(x,y) = 0 \text{ at the lower-left of the joint range.}$$

If the lower-left corner is $(-\infty, -\infty)$, then

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x,y) = 0$$

L15_L17

Property 3

$$F(x,y) = 1 \text{ at the upper-right of the joint range.}$$

If the upper-right corner is (∞, ∞), then

$$\lim_{(x,y)\to(\infty,\infty)} F(x,y) = 1$$

L15_L17

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11. End of Lecture

The final slide is a “Thank You!” slide.

L15_L17

If you want, I can also convert this exact scribe into clean LaTeX without changing any content.

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Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture L15_16_17_18: Joint Random Variables + Joint Distributions + Examples

The lecture covers the following outline: Why Study Joint Random Variables; Discrete Case; Joint PMF Table Representation; Example: Rolling Two Dice; Continuous Case; Geometric Interpretation; Worked Example; Joint Cumulative Distribution Function; Definition of Joint CDF; Continuous Case; Discrete Case; Properties of the Joint CDF; Comprehensive Example. L15_L17

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1. Why Study Joint Random Variables?

In science and in real life, we are often interested in **two (or more) random variables at the same time**. L15_L17

Examples listed

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members L15_L17

Example #1: Smart Transportation

In a smart traffic system, two random variables are defined:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question stated:

- Can speed be high when the traffic density is also high?

Answer shown:

- Yes / No

Conclusion stated:

- So, X and Y are correlated random variables.

Real ITS applications listed:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing L15_L17

Example #2: Wireless Network Performance

In a wireless system, two important random variables are:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Reason stated for joint random variables:

- Throughput depends on SNR
- Both vary due to fading, interference, and congestion

Question stated:

- If SNR is high, will throughput always be high?

Answer shown:

- Yes / No

Reasoning stated:

- Network Congestion (condition relationships)

Network applications listed:

- 5G scheduling
- adaptive modulation
- network planning L15_L17

Example #3

- Weather Prediction $(X = \text{Rainfall and } Y = \text{Temperature})$
- Applications in weather forecasting L15_L17

Example #4

- Drone Delivery $(X = \text{Wind Speed and } Y = \text{Delivery Time})$
- Applications in logistics and UAV systems L15_L17

Example #5

- Roll two dice $(X \text{ and } Y)$ using phones
- $(X + Y = 7 \text{ and } X > Y)$
- This requires joint PMF

L15_L17

Statements concluding the motivation

In such situations, the random variables have a **joint distribution** that allows us to compute probabilities of events involving both variables and understand the **relationship** between the variables. This is simplest when the variables are **independent**. When they are not, we use **covariance** and **correlation** as measures of the nature of the dependence between them.

L15_L17

---

2. Discrete Case – Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X \in \{x_1, x_2, \ldots, x_n\}, \qquad Y \in \{y_1, y_2, \ldots, y_m\}$$

The ordered pair (X, Y) takes values in the product set:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

L15_L17

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

This gives the probability of the joint outcome occurring simultaneously.

L15_L17

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3. Joint PMF Table Representation

The joint PMF is organized in a table. The slide shows rows indexed by values $x_1, x_2, \ldots, x_n$ and columns indexed by values $y_1, y_2, \ldots, y_m$, with table entries of the form $p(x_i, y_j)$.

L15_L17

Key Properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

L15_L17

---

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.
L15_L17

Let

- $X =$ value on the first die
- $T =$ total (sum) of both dice
L15_L17

Since each outcome (i, j) has probability $\frac{1}{36}$, we construct the joint probability table of (X, T).
L15_L17

Joint Probability Table of (X, T)

$X \backslash T$	2	3	4	5	6	7	8	9	10	11	12
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	0
2	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0
3	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0
4	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0
5	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0
6	0	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

L15_L17

Observations

- Entries are either 0 or $\frac{1}{36}$
- X and T are **not independent**
L15_L17

---

5. Continuous Case – Joint PDF

The lecture states that the continuous case is analogous to the discrete case:

- Discrete values $\rightarrow$ Continuous intervals
- Joint PMF $\rightarrow$ Joint PDF
- Sums $\rightarrow$ Double integrals
L15_L17

If

$$X \in [a, b], \qquad Y \in [c, d],$$

then the pair (X, Y) takes values in the product region:

$$[a, b] \times [c, d]$$

L15_L17

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** of X and Y if

$$P\big((X, Y) \text{ lies in a small rectangle of area } dx\,dy\big) \approx f(x, y)\,dx\,dy$$

L15_L17

The geometric interpretation shown is a rectangle of area $dx\,dy$ inside the product region, with probability labeled as:

$$\text{Prob.} = f(x, y)\,dx\,dy$$

L15_L17

Key Properties

$$f(x, y) \geq 0$$

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$$

L15_L17

Additional statement on the slide:

- The joint PDF $f(x, y)$ may take values greater than 1.
- It is a probability density, not a probability.

L15_L17

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6. Worked Example

Given

$$f(x, y) = 4xy, \qquad 0 \leq x \leq 1,\ 0 \leq y \leq 1$$

Tasks

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$

Jointly, (X, Y) takes values in the unit square:

$$[0, 1]^2$$

L15_L17

The slide shades the region corresponding to $A = \{X < 0.5,\ Y > 0.5\}$ in the unit square.

L15_L17

Example Solution

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left(\int_0^1 4xy\,dx \right) dy = \int_0^1 2y\,dy = [y^2]_0^1 = 1$$

Hence $f(x, y) \geq 0$ and total probability is 1. So it is a **valid joint PDF**. L15_L17

Step 2: Compute $P(A)$

$$\begin{aligned} P(A) &= \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx \\ &= \int_0^{0.5} [2xy^2]_{0.5}^1\,dx \\ &= \int_0^{0.5} 2x(1 - 0.25)\,dx \\ &= \int_0^{0.5} \frac{3}{2}x\,dx \\ &= \frac{3}{16} \end{aligned}$$

L15_L17

---

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables. L15_L17

We use the notation:

$$X \leq x,\ Y \leq y$$

to mean the event:

$$\{X \leq x \text{ and } Y \leq y\}$$

L15_L17

Definition

The joint cumulative distribution function is defined as

$$F(x, y) = P(X \leq x,\ Y \leq y)$$

L15_L17

---

8. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, then the joint CDF is obtained by integrating the density:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

L15_L17

Recovering the Joint PDF

Since there are two variables, we use partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x\, \partial y} F(x, y)$$

L15_L17

---

9. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then the joint CDF is obtained by summing probabilities:

$$F(x, y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

L15_L17

Interpretation

- Add all probabilities in the table
- For rows with $x_i \le x$
- And columns with $y_j \le y$ L15_L17

---

10. Properties of the Joint CDF

The joint CDF $F(x, y)$ of X and Y must satisfy several properties. L15_L17

Property 1

$$F(x, y) \text{ is non-decreasing.}$$

If x or y increase, then $F(x, y)$ must stay constant or increase. L15_L17

Property 2

$$F(x, y) = 0 \text{ at the lower-left of the joint range.}$$

If the lower-left corner is $(-\infty, -\infty)$, then

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x,y) = 0$$

L15_L17

Property 3

$$F(x,y) = 1 \text{ at the upper-right of the joint range.}$$

If the upper-right corner is (∞, ∞), then

$$\lim_{(x,y)\to(\infty,\infty)} F(x,y) = 1$$

L15_L17

---

11. End Slide

The final slide contains:

- Thank You!

L15_L17

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Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture L15_16_17_18: Joint Random Variables + Joint Distributions + Examples

L15_L17

Outline

The lecture outline lists the following topics:

- Why Study Joint Random Variables?
- Discrete Case
  - Joint PMF Table Representation
  - Example: Rolling Two Dice
- Continuous Case
  - Geometric Interpretation
  - Worked Example
- Joint Cumulative Distribution Function
  - Definition of Joint CDF
  - Continuous Case
  - Discrete Case
  - Properties of the Joint CDF
- Comprehensive Example

L15_L17

---

1. Why Study Joint Random Variables?

In science and in real life, we are often interested in **two (or more) random variables at the same time**.
L15_L17

Examples

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members
L15_L17

Example #1: Smart Transportation

In a smart traffic system, two random variables are defined:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question stated:

- Can speed be high when the traffic density is also high?

Answer shown:

- Yes / No

Statement given:

- So, X and Y are correlated random variables.

Real ITS Application (probabilistic analysis):

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing
L15_L17

Example #2: Wireless Network Performance

In a wireless system, two important random variables are:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Why Joint Random Variables?

- Throughput depends on SNR
- Both vary due to fading, interference, and congestion

Question stated:

- If SNR is high, will throughput always be high?

Answer shown:

- Yes / No

Reasoning stated:

- Network Congestion (condition relationships)

Network Applications:

- 5G scheduling
- adaptive modulation
- network planning L15_L17

Example #3

Weather Prediction:

- $X =$ Rainfall
- $Y =$ Temperature

Application:

- weather forecasting L15_L17

Example #4

Drone Delivery:

- $X =$ Wind Speed
- $Y =$ Delivery Time

Applications:

- logistics
- UAV systems L15_L17

Example #5

- Roll two dice $(X \text{ and } Y)$ using phones
- $(X + Y = 7 \text{ and } X > Y)$
- This requires joint PMF L15_L17

Statements concluding the section

In such situations, the random variables have a **joint distribution** that allows us to compute probabilities of events involving both variables and understand the **relationship** between the variables.

This is simplest when the variables are **independent**.

When they are not, we use **covariance** and **correlation** as measures of the nature of the dependence between them.
L15_L17

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2. Discrete Case – Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X \in \{x_1, x_2, \ldots, x_n\}, \qquad Y \in \{y_1, y_2, \ldots, y_m\}$$

The ordered pair (X, Y) takes values in the product set:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

L15_L17

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

This gives the probability of the joint outcome occurring simultaneously.
L15_L17

---

3. Joint PMF Table Representation

The joint PMF is organized in a table. The table has rows indexed by values of X and columns indexed by values of Y, with entries $p(x_i, y_j)$.
L15_L17

Key Properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

L15_L17

---

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.
L15_L17

Let

- $X =$ value on the first die

- $T =$ total (sum) of both dice

L15_L17

Since each outcome (i, j) has probability $\frac{1}{36}$, we construct the joint probability table of (X, T). L15_L17

Joint Probability Table of (X, T)

$X \backslash T$	2	3	4	5	6	7	8	9	10	11	12
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	0
2	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0
3	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0
4	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0
5	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0
6	0	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

L15_L17

Observations

- Entries are either 0 or $\frac{1}{36}$
- X and T are **not independent** L15_L17

---

5. Continuous Case – Joint PDF

The lecture states that the continuous case is analogous to the discrete case:

- Discrete values $\to$ Continuous intervals
- Joint PMF $\to$ Joint PDF
- Sums $\to$ Double integrals L15_L17

If

$$X \in [a, b], \qquad Y \in [c, d],$$

then the pair (X, Y) takes values in the product region:

$$[a, b] \times [c, d]$$

L15_L17

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** of X and Y if

$$P\big((X, Y) \text{ lies in a small rectangle of area } dx\,dy\big) \approx f(x, y)\,dx\,dy$$

L15_L17

Geometric Interpretation

The region shown is the rectangle $[a, b] \times [c, d]$, and inside it a small rectangle of dimensions dx and dy is marked with probability

$$\text{Prob.} = f(x, y)\, dx\, dy$$

L15_L17

Key Properties

$$f(x, y) \geq 0$$

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

L15_L17

Additional statement given:

- The joint PDF $f(x, y)$ may take values greater than 1.
- It is a probability density, not a probability. L15_L17

---

6. Worked Example

Given

$$f(x, y) = 4xy, \qquad 0 \leq x \leq 1,\ 0 \leq y \leq 1$$

Tasks

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$

Jointly, (X, Y) takes values in the **unit square**

$$[0, 1]^2$$

L15_L17

The figure shows the region A inside the unit square corresponding to $X < 0.5$ and $Y > 0.5$. L15_L17

Example Solution

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy\, dx\, dy = \int_0^1 \left(\int_0^1 4xy\, dx \right) dy = \int_0^1 2y\, dy = [y^2]_0^1 = 1$$

Hence $f(x, y) \geq 0$ and total probability is 1. So it is a **valid joint PDF**.

L15_L17

Step 2: Compute $P(A)$

$$
\begin{aligned}
P(A) &= \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx \\
&= \int_0^{0.5} [2xy^2]_{0.5}^{1} \, dx \\
&= \int_0^{0.5} 2x(1 - 0.25) \, dx \\
&= \int_0^{0.5} \frac{3}{2} x \, dx \\
&= \frac{3}{16}
\end{aligned}
$$

L15_L17

---

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

L15_L17

We use the notation:

$$X \leq x, \; Y \leq y$$

to mean the event:

$$\{X \leq x \text{ and } Y \leq y\}$$

L15_L17

Definition

The joint cumulative distribution function is defined as

$$F(x, y) = P(X \leq x, \; Y \leq y)$$

L15_L17

---

8. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, then the joint CDF is obtained by integrating the density:

$$F(x, y) = \int_c^y \int_a^x f(u, v) \, du \, dv$$

L15_L17

Recovering the Joint PDF

Since there are two variables, we use partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

L15_L17

---

9. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then the joint CDF is obtained by summing probabilities:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

L15_L17

Interpretation

- Add all probabilities in the table
- For rows with $x_i \leq x$
- And columns with $y_j \leq y$ L15_L17

---

10. Properties of the Joint CDF

The joint CDF $F(x, y)$ of X and Y must satisfy several properties. L15_L17

Property 1

$$F(x, y) \text{ is non-decreasing.}$$

If x or y increase, then $F(x, y)$ must stay constant or increase. L15_L17

Property 2

$$F(x, y) = 0 \text{ at the lower-left of the joint range.}$$

If the lower-left corner is

$$(-\infty, -\infty),$$

then

$$\lim_{(x,y) \to (-\infty, -\infty)} F(x, y) = 0.$$

L15_L17

Property 3

$$F(x, y) = 1 \text{ at the upper-right of the joint range.}$$

If the upper-right corner is

$$(\infty, \infty),$$

then

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1.$$

L15_L17

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11. End Slide

The final slide contains:

$$\text{Thank You!}$$

L15_L17
Sources

You are generating a lecture scribe intended to serve as formal reference material for closed-notes exam preparation.

Use ONLY the attached lecture slides / lecture PDF as context. Do NOT use any textbook or external knowledge.

The scribe must be a faithful reconstruction of what was taught in class. Every definition, notation, assumption, result, proof step, derivation step, and example step must be directly traceable to the provided slides.

While writing the scribe, explicitly reason step by step through:

Definitions and notation

Assumptions and conditions

Statements of results or theorems

Proofs or derivations (only where they appear in the slides)

Worked examples (only where they appear in the slides), including all intermediate steps

Make all logical dependencies between concepts explicit so that a student can reliably revise using only this document.

Follow the lecture's original structure and scope. Preserve ordering of topics as presented.

Maintain a formal academic tone suitable for exam revision.

STRICT CONSTRAINTS:

Do NOT introduce intuition, interpretations, simplifications, or additional explanations.

Do NOT add new examples, proofs, or derivations.

Do NOT infer missing steps unless they are explicitly shown in the slides.

Do NOT use background knowledge.

Do NOT invent content.

This output is a lecture scribe for exam preparation, not a tutorial, summary, or creative explanation.

Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture L15_16_17_18: Joint Random Variables + Joint Distributions + Examples

L15_L17

Outline

The lecture outline is:

- Why Study Joint Random Variables?
- Discrete Case
  - Joint PMF Table Representation
  - Example: Rolling Two Dice
- Continuous Case
  - Geometric Interpretation
  - Worked Example

- Joint Cumulative Distribution Function
  - Definition of Joint CDF
  - Continuous Case
  - Discrete Case
  - Properties of the Joint CDF
- Comprehensive Example L15_L17

---

1. Why Study Joint Random Variables?

In science and in real life, we are often interested in **two (or more) random variables at the same time**. L15_L17

Examples

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members L15_L17

Example #1: Smart Transportation

In a smart traffic system, two random variables are defined:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question stated:

- Can speed be high when the traffic density is also high?

Answer shown:

- Yes / No

Statement given:

- So, X and Y are correlated random variables.

Real ITS Application (probabilistic analysis):

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing L15_L17

Example #2: Wireless Network Performance

In a wireless system, two important random variables are:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Why Joint Random Variables?

- Throughput depends on SNR
- Both vary due to fading, interference, and congestion

Question stated:

- If SNR is high, will throughput always be high?

Answer shown:

- Yes / No

Reasoning:

- Network Congestion (condition relationships)

Network Applications:

- 5G scheduling
- adaptive modulation
- network planning L15_L17

Example #3

Weather Prediction:

- $X =$ Rainfall
- $Y =$ Temperature

Application:

- weather forecasting L15_L17

Example #4

Drone Delivery:

- $X =$ Wind Speed
- $Y =$ Delivery Time

Applications:

- logistics

- UAV systems

L15_L17

Example #5

- Roll two dice $(X \text{ and } Y)$ using phones
- $(X + Y = 7 \text{ and } X > Y)$
- This requires joint PMF

L15_L17

Statements concluding the section

In such situations, the random variables have a **joint distribution** that allows us to compute probabilities of events involving both variables and understand the **relationship** between the variables.

This is simplest when the variables are **independent**.

When they are not, we use **covariance** and **correlation** as measures of the nature of the dependence between them.

L15_L17

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2. Discrete Case – Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X \in \{x_1, x_2, \ldots, x_n\}, \qquad Y \in \{y_1, y_2, \ldots, y_m\}$$

The ordered pair (X, Y) takes values in the product set:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

L15_L17

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

This gives the probability of the joint outcome occurring simultaneously.

L15_L17

---

3. Joint PMF Table Representation

We organize the joint PMF in a table. The table has rows indexed by values of X and columns indexed by values of Y, with entries $p(x_i, y_j)$.

L15_L17

Key Properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n}\sum_{j=1}^{m} p(x_i, y_j) = 1$$

L15_L17

---

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.
L15_L17

Let

- $X =$ value on the first die
- $T =$ total (sum) of both dice
L15_L17

Since each outcome (i, j) has probability $\frac{1}{36}$, we construct the joint probability table of (X, T).
L15_L17

Joint Probability Table of (X, T)

$X \backslash T$	2	3	4	5	6	7	8	9	10	11	12
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3	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0
4	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0
5	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0
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L15_L17

Observations

- Entries are either 0 or $\frac{1}{36}$
- X and T are **not independent**
L15_L17

---

5. Continuous Case – Joint PDF

The continuous case is analogous to the discrete case:

- Discrete values $\longrightarrow$ Continuous intervals
- Joint PMF $\longrightarrow$ Joint PDF
- Sums $\longrightarrow$ Double integrals
L15_L17

If

$$X \in [a, b], \qquad Y \in [c, d],$$

then the pair (X, Y) takes values in the product region:

$$[a, b] \times [c, d]$$

L15_L17

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** of X and Y if

$$P\big((X, Y) \text{ lies in a small rectangle of area } dx\,dy\big) \approx f(x, y)\,dx\,dy$$

L15_L17

Geometric Interpretation

The figure shows the product region $[a, b] \times [c, d]$ and a small rectangle inside it with dimensions dx and dy, labeled with probability

$$\text{Prob.} = f(x, y)\,dx\,dy$$

L15_L17

Key Properties

$$f(x, y) \geq 0$$

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$$

L15_L17

Additional statement:

- The joint PDF $f(x, y)$ may take values greater than 1.
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L15_L17

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6. Worked Example

Given

$$f(x, y) = 4xy, \qquad 0 \leq x \leq 1,\ 0 \leq y \leq 1$$

Tasks

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

$$A = \{X < 0.5,\ Y > 0.5\}$$

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Jointly, (X, Y) takes values in the unit square

$$[0,1]^2$$

L15_L17

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Example Solution

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left(\int_0^1 4xy\,dx \right) dy = \int_0^1 2y\,dy = [y^2]_0^1 = 1$$

L15_L17

Hence $f(x, y) \geq 0$ and total probability is 1. So it is a **valid joint PDF**. L15_L17

Step 2: Compute $P(A)$

$$\begin{aligned} P(A) &= \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx \\ &= \int_0^{0.5} [2xy^2]_{0.5}^1\,dx \\ &= \int_0^{0.5} 2x(1 - 0.25)\,dx \\ &= \int_0^{0.5} \frac{3}{2}x\,dx \\ &= \frac{3}{16} \end{aligned}$$

L15_L17

---

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Suppose X and Y are jointly distributed random variables. L15_L17

We use the notation

$$X \leq x,\ Y \leq y$$

to mean the event

$$\{X \leq x \text{ and } Y \leq y\}$$

L15_L17

Definition

The joint cumulative distribution function is defined as

$$F(x, y) = P(X \leq x,\ Y \leq y)$$

L15_L17

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If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, then the joint CDF is obtained by integrating the density:

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L15_L17

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L15_L17

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L15_L17

Interpretation

- Add all probabilities in the table
- For rows with $x_i \leq x$
- And columns with $y_j \leq y$ L15_L17

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The joint CDF $F(x, y)$ of X and Y must satisfy several properties. L15_L17

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$$F(x, y) \text{ is non-decreasing.}$$

If x or y increase, then $F(x, y)$ must stay constant or increase.

L15_L17

Property 2

$$F(x, y) = 0 \text{ at the lower-left of the joint range.}$$

If the lower-left corner is

$$(-\infty, -\infty),$$

then

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

L15_L17

Property 3

$$F(x, y) = 1 \text{ at the upper-right of the joint range.}$$

If the upper-right corner is

$$(\infty, \infty),$$

then

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

L15_L17

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11. End Slide

$$\text{Thank You!}$$

L15_L17

Sources